

Mashalakh Mamedov

Mechatronics Engineering MSc Student | BSc Electrical and Electronics Engineering | Power Electronics, MATLAB and Simulink Modelling and Validation

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Available from October 2026, 6 to 12 months

MSc candidate in Mechatronics Engineering at the University of Rome Tor Vergata, with a BSc in Electrical and Electronics Engineering. I model and validate electromechanical systems in MATLAB, Simulink, Simscape Electrical and Autodesk Inventor: power conversion, robot dynamics, vehicle energy and image-based evaluation, in the same toolchain. Two years nine months of prior on-site industrial experience on a gas-pipeline expansion project. Seeking a 6 to 12 month engineering internship from October 2026 in power electronics or adjacent controls, model-based design, robotics or test and validation.

EDUCATION

MSc, Mechatronics Engineering · University of Rome Tor Vergata, Rome, Italy

Sep 2024 to expected 2027

Taught in English. Currently enrolled, second year.

- Coursework: Power Electronics and Electrical Drives, Robot Mechanics, Computer Vision, Integrated Sensors, VLSI Circuit and System Design, Fundamentals of Mechanics of Systems, Thermodynamics and Heat Transfer.

BSc, Electrical and Electronics Engineering · Tbilisi State University

Awarded Jul 2023

Ivane Javakhishvili Tbilisi State University, Tbilisi, Georgia. GPA 2.9 / 4.0. Senior Design Project: A (96).

SELECTED ENGINEERING PROJECTS

3 kVA Single-Phase H-Bridge Inverter

A.Y. 2025/26

MSc Power Electronics and Electrical Drives, Tor Vergata. Individual project. Simulation only, with no prototype, PCB or laboratory measurement.

- Designed and simulated a 3 kVA single-phase H-bridge inverter, 400 V DC to 230 V RMS at 50 Hz with unipolar SPWM and a 5 kHz carrier, as a parameterised MATLAB and Simulink model in Simscape Electrical Specialized Power Systems.
- Sized a 5 mH / 20 μ F LC output filter and validated across unity, 0.5 inductive and 0.5 capacitive power factors plus no-load and full-load step transients, reaching **1.64% output voltage THD** at full load against a below-3% requirement.
- Diagnosed an LC filter resonance under inductive loading, where the load current stayed smooth while the voltage rang, and cut THD from 3.64% to **2.33%** with a 5 Ω series damping resistor, reporting the resulting 0.72% to 1.64% rise at unity power factor as the cost of the fix.

Fire and Smoke Semantic Segmentation

Jun to Jul 2026

MSc Computer Vision, Tor Vergata. Two-person project; I owned the deep-learning workstream end to end.

- Built the dataset pipeline recovering binary masks from COCO polygon annotations, then trained DeepLabv3+ on a ResNet-18 backbone with transfer learning and inverse-frequency class weighting across a 141 / 40 / 20 split.
- Benchmarked the network against a teammate's classical colour-threshold and morphology baseline on identical test data and metric, reaching a **mean test F-measure of 0.8507 against 0.4421**.
- Traced a NaN mean F-measure to zero-true-positive images, which led to auditing the labels: one test image showing clear fire carried an empty reference mask, worth about **4.48 percentage points** of the test mean. Kept it in the results and reported it as a data-quality finding.

RoboLatte: UR5e Dynamic Simulation

Jan 2026

MSc Robot Mechanics, Tor Vergata. Two-person project; I owned the Autodesk Inventor workstream. Simulation and CAD only.

- Built the UR5e CAD assembly in Autodesk Inventor, drove all six revolute joints with imposed cubic ramp motions, and configured Dynamic Simulation with gravity, CAD-derived inertias and a 3 N end-effector payload.
- Extracted joint driving forces from the Output Grapher at three instants and unit-converted N·mm to N·m, deriving an offset joint-convention mapping so the results were comparable with my teammate's analytical MATLAB inverse dynamics.
- Five of six joints agreed in sign and order of magnitude; Joint 3 did not, traced to the approximate link inertias the MATLAB model was forced to use. Reported rather than smoothed over.

V2V-Inspired Eco-Driving Energy Analysis

2026

MSc Powertrain Technologies for Future Mobility, Tor Vergata. Two-person project; I owned the MATLAB and Simulink vehicle model. Simulation only.

- Built a longitudinal road-load and traction-energy model with a parallel Simulink block implementation, quantifying a **9.7% traction-energy saving** (881.5 to 795.6 kJ; 8.47 to 7.62 L/100 km) from an anticipating speed profile.
- Reconciled script and block model to 0.0% discrepancy after tracing the mismatch to central versus one-sided differentiation and aligning the solver. This is an implementation-consistency check, not independent validation.

Remotely Operated Mobile Manipulator

2023

BSc Senior Design Project, Tbilisi State University. Four-person team; individual task split not documented.

- Led and co-authored the technical report and contributed across Arduino-based control, system integration and troubleshooting for a commercial 4-axis arm integrated onto a DIY 4WD platform with an NRF24L01 2.4 GHz link, L298N drive electronics and a 3D-printed gripper.
- The team diagnosed motor-induced supply sag causing Arduino resets and resolved it by separating the motor and logic power domains.

EXPERIENCE

Mechanical Installation Support Technician & Interpreter · Bechtel-Enka JV	Feb 2016 to Jul 2017
<i>South Caucasus Pipeline Expansion (SCPX). Gardabani, Georgia, on site.</i>	
• Worked from P&ID schematics and technical drawings on an active gas-pipeline expansion site, assisting with industrial machinery alignment during installation and shutdown work. • Interpreted on site between Siemens, Solar Turbines and Ingersoll Rand vendor engineers and Georgian crews, and tracked critical components through shutdown work under BP and Bechtel EHS standards.	
Warehouse Data Entry Clerk · Bechtel-Enka JV	Jul 2017 to Oct 2018
<i>Same SCPX project. Gardabani, Georgia. Two years nine months on site in total.</i>	
Data Annotation Specialist · ALLSTARSIT	Jun 2024 to Oct 2024
<i>Tbilisi, Georgia.</i>	
• Annotated autonomous-driving imagery in a proprietary Linux-based tool, covering 2D and 3D tasks for vehicles, pedestrians, traffic lights and road markings, to predefined annotation policies with QA and team-lead review workflows.	

TECHNICAL SKILLS

Modelling & simulation	MATLAB · Simulink · Simscape Electrical Specialized Power Systems · parameterised model design · solver selection · script-versus-model reconciliation
Power electronics	H-bridge inverter design · unipolar SPWM · LC filter design and damping · FFT/THD analysis · power-factor load modelling · semiconductor selection with loss and thermal estimation
Robotics & control	Forward and inverse kinematics · Denavit-Hartenberg modelling · workspace analysis · trajectory planning · inverse dynamics · CAD dynamic simulation
Computer vision	Semantic segmentation · DeepLabv3+ · ResNet-18 · transfer learning · class weighting · COCO-to-mask dataset preparation · F-measure evaluation · label-quality auditing
Automotive	Longitudinal vehicle dynamics · road-load and traction-energy modelling · model verification
Embedded & CAD	Arduino / C++ · NRF24L01 2.4 GHz · servo control · L298N motor drivers · Autodesk Inventor

LANGUAGES, CERTIFICATION AND ADDITIONAL EXPERIENCE

Languages	Turkish (native) · Azerbaijani (native) · English (C1 working proficiency) · Georgian (B2) · Russian (B1) · Italian (basic)
Certification	EF SET English Certificate: 74/100, C2 Proficient (Reading 75, Listening 72), awarded 30 January 2025
Additional	iGaming Customer Support Specialist, RISK, Remote / Rome · May 2025 to present · Customer Service Specialist, Majorel, Tbilisi · Feb 2022 to Jun 2024